Macroeconomics of Risks

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Abstract

This paper studies the impact of a government spending shock on the economy through key macroeconomic indicators, which are the Gross Domestic Product and consumption. We also consider the Economic Policy Uncertainty index, which measures the uncertainty regarding government economic policies. To do so, a Vector Auto-Regressive (VAR) model is fitted through a recursive Cholesky decomposition. Impulse Response Functions (IRF) are used to study the dynamic propagation of fiscal shocks. The results show that government spending shocks generate medium-run fluctuations (around 5 quarters) in GDP and consumption before returning to baseline. However, the uncertainty index presents limited response. This suggests that fiscal policy has transitory real effects on the economy but a little impact on economic uncertainty.

Keywords: Shock Propagation, Vector Auto-Regressive, Impulse Response Function

Contents

1 Introduction

Macroeconomic risk is a characteristic of modern economies. On top of the recent disasters that have hit the economy such as the GFC and Covid-19 crisis, the daily life of agents is characterized by small and more frequent fluctuations, also known as Business as Usual (BaU) shocks. Understanding the nature and transmission of these shocks is central to macroeconomic analysis and policy design.

To better understand the nature of these fluctuations, we clarify what we mean by "risk" in this report. In 1931, Knight highlighted the difference between risk and uncertainty. According to his words, risk is a situation under which possible realizations of future events and their probabilities of occurrences are known to the decision-maker. Uncertainty is the situation under which such (either one or both) information is not available to the decision-maker.

BaU and Disaster shocks can be explained according to two main criteria: Frequency and Size. For BaU, shocks are observed frequently, so there is enough information to estimate them. Concerning Disaster, there is not much information to estimate, so it is more difficult to comprehend. The size of BaU shocks is moderate and call for stabilization policies but in reality they may not be very harmful. On the contrary, Disasters are much larger and may call for extreme policies, also, they are potentially very harmful. In this analysis, we concentrate on measurable risks, with exogenous shocks and Business-as-Usual fluctuations.

The contribution of this study lies in the empirical analysis of fiscal policy shocks and their dynamic effects on key macroeconomic variables. Using a structural Vector Autoregressive (SVAR) framework, we examine the responses of output, consumption, and government spending to a government spending shock. However, the main contribution of this work concerns the Economic Policy Uncertainty (EPU) index, which is used as a proxy for policy related uncertainty. By incorporating the EPU index into the VAR system, this report assesses whether fiscal shocks significantly affect measured economic policy uncertainty.

Structural identification is achieved through a Cholesky decomposition of the reduced-form VAR residuals, following the approach of Blanchard and Perotti (2002). This framework allows us to asses the dynamic propagation of fiscal shocks and evaluate their relative importance using impulse response functions and forecast error variance decompositions.

2 Data description

2.1 Sources and period

For the empirical analysis, two data sets are used. The first one is a quarterly database from 1947 to the second quarter of 2015, this accounts for 275 time observations of nominal macroeconomic variables. The first dataset includes three main types of variables:

- Nominal variables
- Deflators
- Other general variables

Nominal variables contain government data such as nominal GDP, nominal total consumption, nominal durable consumption expenditures, nominal nondurable consumption, nominal services consumption, etc ... The databae contains also deflators. A deflator is a price index used to remove the effect of inflation from an economic variable expressed in monetary term, such as GDP or consumption deflator. Finally, there are some general variables, for instance total population and 3-month Treasury bill.
We are interested only in a selection of these variables that we summarize in Table 2.

The second database contains monthly data from January 1900 to October 2025 on the news based policy uncertainty index (or Economic Policy Uncertainty index, EPU).

The frequency of the two databases are not the same. Indeed, the first one has a quarterly frequency while the second one, has monthly frequency. We take the 3 month average of the monthly data to have only quarterly data, by doing that we align the data on the same frequency. In addition, we only keep the periods covered by the two databases. At the end, we have a quarterly database ranging from 1947 to the second quarter of 2015. We highlight that it has been chosen to work with quarterly data for consistency with the dynamics studied in VAR models presented in the second part of the class.

2.2 Economic justification of variables used

For this study, we are interested in the variables commonly used in standard macroeconomics models. Indeed, our variables of interest are:

- **Real GDP** is used as a measure of aggregate output, which corresponds, for example, to the equilibrium level of production in standard DSGE models.
- **Real consumption** reflects intertemporal choices made by households. It captures welfare and risk-sharing decisions.

- **Government spending** is treated as an exogenous policy variable, in line with standard macroeconomic models, and is commonly used to identify fiscal shocks. (cf. Blanchard & Perotti, Gali et al.)
- The **Economic Policy Uncertainty index** developed by Baker, Bloom, and Davis (2016) measures policy-related uncertainty. This index captures perceived macroeconomic risk and expectations, which can influence economic decisions even in the absence of realized macroeconomic shocks.

Hence we need to transform the nominal variables of the database into real ones to conduct properly our analysis.

3 Variables preparation

3.1 Real variables

We work with real variables in order to isolate fluctuations in economic activity from movements in the general price level. Using nominal variables would confound real effects with inflationary dynamics and complicate the identification of macroeconomic shocks. The selected nominal variables are then converted into real variables by deflating them using the corresponding deflators.

The following conversions are applied:

- Real GDP : $Y_t = \frac{ngdp_t}{pgdp_t}$
- Real consumption (non-durable goods and services) : $C_t = \frac{(ncnd+ncsv)_t}{pcndsv_t}$
- Real government spending : $G_t = \frac{ngov_t}{pgov_t}$

Furthermore, consumption is adjusted to exclude durable goods in order to focus on current consumption of services and non-durable goods, which better reflects the immediate behavior of households.

3.2 Logarithmic transformation

We take the logarithm of the macroeconomic variables, transforming multiplicative growth into additive variations and stabilizing variance. Indeed, since most of macroeconomic variables have exponential growth, taking logarithm stabilizes them by making them linear. This insures homoskedasticity, which is required for time series estimation. By doing so, our model now aligns with log linearized model (such as DGSE) and we obtain coefficients that can be interpreted as percentages. This step will facilitate stationarity and econometric analysis.

3.3 Non-stationarity problem

The VAR framework assumes that the vector of observed macroeconomic variables follows a stable multivariate auto-regressive process. Therefore, before estimation, it is necessary to ensure that the variables are stationary, so that the VAR admits a valid Wold representation, and impulse response functions are well defined.

In macroeconomic time series, non-stationarity often arises from stochastic trends, typically associated with the presence of a unit root. In such cases, shocks have permanent effects on the level of the series. Estimating a VAR with non-stationary variables may lead to false regressions, where statistically significant relationships arise despite the absence of any meaningful economic link.

Therefore, identifying and removing non-stationary components is an important step for valid inference in our multivariate time series model.

3.4 Stationarization methods

Stationarity implies that the statistical properties of the series namely their mean, variance, and autocovariance structure are constant over time. However, the macroeconomic variables considered in this study, such as real consumption (Figure 1), GDP (Figure 2), and government spending (Figure 3), exhibit clear upward trends, (observed graphically for now), over the sample period and are confirming non-stationarity.

Because we focus on business-cycle fluctuations rather than long-run growth, these series must be transformed into stationary real variables. Two standard approaches are commonly used in the literature

- Removing deterministic trends
- Taking first differences

Both methods aim to eliminate non-stationary components while preserving the relevant cyclical dynamics of the variables.

3.4.1 Linear detrending approach

This first approach, based on Blanchard–Perotti (2002), consists in modeling the trend as a linear function of time:

$$y_t = \alpha + \beta t + \varepsilon_t$$

Where α and β capture the linear trend over time. The residuals represent the deviations from this trend: the short-term fluctuations we are interested in.
Then, we estimate the trend using linear regression:

- First, we fit three linear models with the variables (Y_t, C_t and G_t) as the dependent variables and time as the explanatory variable
- Second, for each of the three models, we subtract the trend (fitted values) from the original series, in order to obtain the detrended series

The residuals are then the detrended series, which fluctuates around zero and can be treated as approximately stationary for further analysis. The trends and cyclical fluctuations detrended are illustrated for the three macroeconomic variables in Figure 4 to Figure 6.

3.4.2 First difference approach

The second method we consider for making our time series stationary is to use first differences from Gali (1992), i.e., the differences between consecutive values in the series:

$$\Delta y_t = y_t - y_{t-1}$$

This transformation eliminates linear or exponential trends and stabilizes the mean of the series. Applying first differences to logarithmic series is equivalent to computing the growth rates of a variable, which makes the fluctuations proportional to the level of the series and facilitates interpretation.
The series in first differences are represented in Figure 7 to Figure 9. We clearly see that they are all three stationary.

These two methods, the linear detrending and the first-difference approaches, isolate the short term fluctuations of variables around its long-term linear trend, allowing us to focus on the dynamics of interest without being influenced by the overall growth trend.

The choice of one of the method between the two has economic implications. Indeed, linear detrending assumes that shocks generate temporary deviations from a deterministic long-run path, which is consistent with business-cycle models where the economy reverts to a balanced growth trajectory. By contrast, first differencing treats shocks as having permanent effects on the level of the variables, emphasizing growth-rate dynamics rather than level fluctuations.

To determine the appropriate transformation for each macroeconomic variable, we perform unit root tests on the logarithms of the real series. Specifically, we rely on the Augmented Dickey–Fuller (ADF) test to assess whether the series are stationary. We allow for a deterministic time trend in the test specification, which is appropriate for macroeconomic variables that typically exhibit long-run growth.

The null hypothesis of the test is the presence of a unit root. The alternative hypothesis corresponds to stationarity around a deterministic trend. The lag length of 4 is chosen to account for serial correlation in the residuals, ensuring the validity of the test.

These tests allow us to distinguish between stochastic and deterministic trends and help choose whether variables should be first-differenced or detrended prior to VAR estimation. The results of the tests are the following.

Variable	ADF statistic
GDP	-1.13
Consumption	-0.0038
Public spending (G_t)	-4.0952

Table 1: Augmented Dickey-Fuller unit root tests

- For the **GDP**, we have $-1.13 > -3.13$. The null hypothesis is not rejected even at 10%. Hence, the series has a unit root. It is not trend-stationary. It is integrated of order 1 (I(1)).
- For **consumption**, $-0.0038 > -3.13$. The null hypothesis is not rejected even at 10%. Therefore, consumption is not stationary around a deterministic trend. It has a stochastic trend. It is I(1).
- For **public spending**, $-4.0952 < -3.98$. The null hypothesis is rejected at 1%. Unlike for GDP and consumption, log(Gt) does not have a unit root. The series is stationary around a deterministic trend. Therefore, Gt is trend-stationary (TS).

To conclude, the ADF tests with a deterministic trend indicate that real GDP and real consumption are integrated of order one, while real government spending appears to be trend stationary (TS). Since we have to choose a common de-trending method for all the three variables, we transform the log-level real variables by taking first differences (2 I(1) against 1 TS), insuring an homogeneous structure.
In addition to real macroeconomic variables, the empirical analysis includes **the economic policy uncertainty index**, no transformations are applied to it.
We define:

- $y_t = \Delta log(Y_t)$
- $c_t = \Delta log(C_t)$

- $g_t = \Delta log(G_t)$
- $i_t = Index$

4 VAR

4.1 VAR motivation

The Vector AutoRegressive (VAR) model is a multivariate extension of the univariate autoregressive framework. It is particularly used for the analysis of macroeconomic dynamics, because it provides a flexible representation of the joint evolution of multiple endogenous variables.

The use of a VAR model is consistent with the theoretical framework introduced in the course. Linearized RBC, New Keynesian and DSGE models imply a system of dynamic equations in which macroeconomic variables respond over time to structural shocks. Unlike DSGE models, however, VAR models impose no a priori restrictions on the dynamic coefficients or on the covariance structure of the innovations. The VAR should be interpreted as a reduced form approximation of equilibrium dynamics rather than a micro-founded structural model.

Moreover, all variables in our system are treated as endogenous, reflecting the presence of dynamic interactions and feedback effects among macroeconomic aggregates. Each variable depends on its own past realizations as well as on the lagged values of the other variables in the system. This framework captures delayed and persistent responses to shocks, as the effects of macroeconomic disturbances (such as government spending shocks in our case) may unfold gradually over time rather than instantaneously.

Once the VAR is structurally identified, this framework allows us to study the dynamic effects of structural shocks through impulse response functions and variance decompositions.

4.2 VAR specification

This section precisely explains how we estimate our VAR model. First of all, there is a trade-off regarding the number of variables to include in it. If too few variables are selected, some dynamic interpretations can be missing, but if there are too many, then there may be a lack of precision, due to the huge number of parameters to estimate. The maximum number to consider is 8. We select in our model 4 variables : the GDP, Non durable Consumption, Government Spending and an Uncertainty Index (EPU : Economic Policy Uncertainty). According to time series theory, the right number of lags should be determined relying on information criteria such as AIC, BIC or HQ. It is important to select the number of lags fitting the best the data. If too few lags are selected, then the model may fail to capture the true dynamic interactions between the variables, and too many lags lead to over parametrization and imprecise estimates. Overall, it is a bias-variance trade-off. In practice, we rely on the AIC criteria that made us choose p=3 for our model.

4.3 Diagnostic tests

Before going further, we have to check that the model is valid. Hence several diagnostic tests are conducted to assess the adequacy of the VAR specification.

First, the multivariate Portmanteau test for serial correlation indicates evidence of residual autocorrelation since we reject the null ($p = 0.042$) at 5%.

Second, the multivariate ARCH test reject the null hypothesis of homoskedastic residuals ($p < 2.2e^{-16}$), implying that conditional heteroskedasticity is a concern in the estimated VAR. Hence the variance of residuals is not constant accross time.

Finally, the multivariate Jarque-Bera test indicates that the residuals are not approximately normally distributed (again H_0 is strongly rejected $p < 2.2e^{-16}$). Even if normality is not strictly required for consistent VAR estimation, it supports the validity of inference based on asymptotic confidence intervals for impulse responses.

Overall, these diagnostic results suggest that the selected lag length is not sufficient to capture the dynamic interactions among variables.

To counter this problem, we choose to work, as recommended during the explanation of the project, with 4 lags instead of 3. If we redo the tests with 4 lags we have now no evidence of residual autocorrelation since the Portmanteau test don't reject anymore the null at 5% level ($p = 0.073$), confirming that residuals are white noises. In addition, the ARCH also shows an improvement with a decrease of the chi-square statistic.

This choice aligns with standard time series theory where it is common to use 4 lags when working with quarterly data. As a consequence we retain a VAR(4), prioritizing the statistical validity of residuals over strict optimization of the AIC criterion.

4.4 The model

We define the vector of variables $Z_t = \begin{pmatrix} y_t \\ c_t \\ g_t \\ i_t \end{pmatrix} \in \mathbb{R}^4$. We consider the following VAR(4) model:

$$Z_t = A_0 + A_1 Z_{t-1} + A_2 Z_{t-2} + A_3 Z_{t-3} + A_4 Z_{t-4} + u_t, \qquad t = 1947, \ldots, 2015, \tag{1}$$

where:

$A_0 \in \mathbb{R}^4$,

$A_j \in \mathbb{R}^{4\times 4}, \quad j = 1, 2, 3, 4$

$u_t \sim \mathcal{N}(0, \Sigma_u)$.

4.5 Identification

In practice, in order to analyze the dynamic effects of a government spending shock, we first estimate a reduced form VAR, using a recursive identification scheme based on a Cholesky decomposition of the variance-covariance of residuals, following Blanchard and Perotti (2002).

The Cholesky decomposition imposes a causal ordering among variables, implying that variables ordered earlier are not affected by shocks to variables ordered later within the same quarter.

Government spending is ordered first in the VAR. This assumption reflects rigidities in the fiscal decision process, as government spending is largely predetermined within a quarter due to budgetary procedures, legislative delays, and implementation lags. Hence, government spending is assumed not to respond to innovations in output, consumption, or economic policy uncertainty.

Output is ordered second, allowing GDP to respond to government spending shocks, implying a feedback from consumption and uncertainty within the same quarter. This assumption is consistent with the idea that aggregate production may adjust rapidly to fiscal impulses, whereas private demand and expectations react with a delay.

Consumption is ordered third. Household consumption decisions are assumed to respond to fiscal and output shocks but not instantaneously to innovations in policy uncertainty. This reflects adjustment frictions, habit formation, and information rigidities.

Finally, the **Economic Policy Uncertainty (EPU) index** is ordered last. This assumption allows the uncertainty index to respond contemporaneously to all macroeconomic shocks in the system, while preventing it from affecting real activity within the same quarter. Given that the EPU index is as a news-based measure, this ordering avoids making wrong assumptions on its causal impact on other variables.

Under this identification choices, the structural government spending shock is interpreted as an exogenous innovation orthogonal to other macroeconomic fluctuations. This framework enables the analysis of impulse response functions and forecast error variance decompositions while maintaining economically interpretable identifying assumptions.

4.6 Impulse Response Function (IRFs)

An Impulse Response Function is a tool measuring the propagation of a shock over time. Considering a variable Y, a shock ε_t at horizon h, it is defined as follows :

$$IRF_h(Y, \varepsilon_t) = \mathbb{E}[Y_{t+h}|\mathcal{I}_t, \varepsilon_t] - \mathbb{E}[Y_{t+h}|\mathcal{I}_t].$$

Where $\mathcal{I}_t$ is the information set available at time t. This concept can be explained by the Rocking Horse model of Ragnar Frisch (1933). In this model, Frisch explains the business cycle as the following : a shock hits the economy (the impulse) and it spreads in the economy (the propagation). The impulse is a short term exogenous, random shock (it can be a government spending, technology shock, ...). This shock then spreads across the economy, persists (but is not permanent) and generates cycles. Here, we are in the case of a shock on Government spending. It has dynamic effects on the GDP, the consumption and the uncertainty (measured here by the EPU index). In that case, if we observe persistent IRFs, this suggests a long propagation of the shock. In the other case, there is a small propagation. Moreover, if the IRFs oscillate, we can suspect cycles.

4.7 Forecast Error Variance Decomposition (FEVD)

As a last step of this analysis, the forecast error variance decomposition is studied. This tool helps us to deepen the understanding of the results given by the IRF. While the IRF shows the dynamic response to a shock, the FEVD shows the importance of each shock.

5 Results

Since individual VAR coefficients are not economically meaningful, we do not report and interpret them in this report. What is really important, is the dynamic propagation of shocks, which is captured by impulse responses and variance decompositions. Since all real variables enter the VAR in log-differences, impulse responses are interpreted as growth-rate responses.

5.1 Response of Government Spending

Result. The response of government spending to its own shock is positive and statistically significant on impact as shown in Table 3 and Figure 10. At horizon zero, the lower and upper bounds of the 95% confidence interval are strictly positive, indicating that the fiscal shock is identified. The response then declines over time and becomes statistically close to zero after approximately 6 to 7 quarters, as the confidence interval starts to include zero.

The results of FEVD (Table 4) shows that fiscal shocks are mostly exogenous. In the long run (quarter 24), 88% of their variance is explained by themselves. For example, GDP shocks explain only 9% of them suggesting the fiscal policy is independant of other variables.

Interpretation. This shows that the shock is transitory and does not permanently affect the growth rate of public spending. The smooth decline and the absence of oscillations support the validity of the Cholesky identification and suggest that the government spending shock captures exogenous innovation rather than endogenous innovation.

5.2 Response of GDP

Result. The Table 5 and Figure 11 show the IRF for GDP. GDP responds positively to a government spending shock on impact but right after the first quarter, this response is not statistically significant anymore at the 95% confidence level, as the confidence interval includes zero at all horizons. While the point estimates suggest a small positive effect in the short run, followed by small fluctuations around zero, the big confidence bands indicate uncertainty around these estimates.

According to the results of the FEVD (Table 6), the variance of GDP is explained, in the long run, by shocks in public spending, by only 6%. GDP explains itself 76% of its variance while consumption explains around 15% showing that public policy has a limited effect on GDP.

Interpretation. From a theoretical perspective, this result is consistent with standard Real Business Cycle and New Keynesian models in which government spending shocks do not necessarily generate strong output responses when financing constraints and intertemporal substitution are taken into account. Public spending stimulates demand, but if households anticipate higher future taxes, they may consume less today, which reduces the overall effect on economic activity.

Moreover, since the VAR is estimated in first differences, impulse responses capture growth-rate effects rather than level effects. This specification may attenuate the estimated persistence of fiscal shocks, leading to smaller and less significant output responses. Overall, the results suggest that government spending shocks do not constitute a dominant source of output fluctuations at business-cycle frequencies.

5.3 Response of Consumption

Result. The response of consumption to a government spending shock is negative on impact, then it is positive and after it is weak and non-monotonic, with small values, and finally converges toward zero over the medium term. This empirical response observed in Table 7 and Figure 12 suggests that Ricardian effects dominate in the short run, while Keynesian demand effects are weak and do not generate a strong consumption effect.

In Table 8 shocks on GDP explain about 23% of the variance consumption in the long run. The reaction of the variance of consumption to its own shocks represents 71% of the total variance and 3% are explained by shocks on government spending.

Interpretation. The results we observe are not consistent with Keynesian theory according to which in the short term, a positive shock in public spending stimulates aggregate demand and therefore production and income. In that case, households can increase their consumption immediately (multiplier effect). This is not what we observe with our IRF. On contrast the negative effect on impact aligns more with the results of Galí, López-Salido, and Vallés (2007), who document that consumption responses to fiscal shocks depend a lot on expectations, fiscal financing, and the monetary policy regime. This pattern is consistent with a framework in which households are forward-looking and partially Ricardian. In standard RBC models for example, a rise in public spending financed by future taxes leads households to reduce current consumption in anticipation of higher future tax liabilities, resulting in low or negative consumption responses.

5.4 Response of Uncertainty

Result. As depicted in Table 9 and Figure 13, on impact, the uncertainty index increases, with an initial response of approximately 0.17, but this effect is never significant since 0 is always in the large confidence interval.

The uncertainty index variance is explained at 99% by its own shocks (Table 10). This is in line with the results shown by the IRF according to which policy-related uncertainty is weakly explained by shocks of the real economy.

Interpretation. This result suggests that fiscal policy shocks do not systematically affect measured policy uncertainty, at least in the short run.

One interpretation is that the EPU index primarily captures news-driven or political sources of uncertainty that are only weakly related to fiscal innovations identified in the VAR. As a result, even well-identified government spending shocks may fail to translate into significant movements in the uncertainty index.

This finding highlights a potential disconnection between fiscal policy actions and perceived policy uncertainty, suggesting that the EPU index may reflect broader institutional or political dynamics rather than short-run macroeconomic shocks.

In line with the conceptual framework presented in Handout 1, this findings highlight the distinction between measurable risk and unmeasurable uncertainty.

6 Conclusion

This report investigates the dynamic effects of government spending shocks on macroeconomic activity and economic policy uncertainty using a structural VAR framework. The analysis focuses on Business-as-Usual fiscal shocks over the period 1947-2015 and examines their transmission to output, consumption, and the Economic Policy Uncertainty (EPU) index.

The results indicate that government spending responds positively and significantly to its own shock, but this effect is transitory and fades after several quarters. Output and consumption responses are weak and generally not statistically significant (0 is included in the confidence intervals), suggesting that fiscal shocks have limited effects on real economic activity in growth rates. Nevertheless, it is worth noting that consumption exhibits an initial negative response, consistent with forward-looking household behavior and Ricardian effects. Regarding uncertainty, the EPU index does not display a statistically significant reaction to government spending shocks (it has very large confidence bands including 0), and its variance is almost entirely explained by its own innovations. These results suggest that, within the present VAR framework, fiscal shocks do not significantly affect the EPU index in the short run.

Several limitations should be acknowledged. First the analysis relies on a linear VAR framework and a recursive identification scheme, which may overestimate non-linear dynamics and alternative transmission mechanisms. Second, our analysis relies on a Cholesky identified SVAR but the recursive order between variables can be discussed. Finally, the scale and volatility of the EPU index differ substantially from real macroeconomic variables, potentially inflating confidence intervals.

Possible extensions of this work can be considered. Non-linearities between variables could be studied and investigated in a further analysis. Then, as explained in the introduction, this report focuses on BaU risks but the same analysis can be conducted looking at Disaster risks following the Barro method to identify them. The SVAR we consider in our analysis is relying on the Cholesky decomposition but alternative identification strategies such as Sign restrictions, Max-Share approach, Narrative approach, DSGE models, Proxy SVARs, High frequency identification, Higher moments, Factor Augmented VAR (FAVAR) can be considered. Finally, for the EPU index, we could have considered a log transformation for example in order to reduce its difference in scale with other variables. These extensions will allow for a deeper understanding of macroeconomic shocks and their propagation in the economy.

References

[1] Blanchard, O. and Perotti, R. (2002). *An empirical characterization of the dynamic effects of changes in government spending and taxes on output*. Quarterly Journal of Economics, 117, 1329–1368.

[2] Canova, F. (2007). *Methods for Applied Macroeconomic Research*. Princeton University Press.

[3] Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). *Nominal rigidities and the dynamic effects of a shock to monetary policy*. Journal of Political Economy, 113(1), 1–45.

[4] Christiano, L. J., Eichenbaum, M., and Vigfusson, R. (2003). *What happens after a technology shock?* National Bureau of Economic Research, Working Paper 9819.

[5] Cogley, T. and Nason, J. M. (2005). *Output dynamics in real business cycle models*. American Economic Review, 95(4), 492–511.

[6] DeJong, D. N. and Dave, C. (2011). *Structural Macroeconometrics*, 2nd edition. Princeton University Press.

[7] Fernald, J. (2026). *US Total Factor Productivity (TFP) Data*. Federal Reserve Bank of San Francisco. Available at: `https://www.frbsf.org/economic-research/indicators-data/total-factor-productivity-tfp/` (Accessed January 2026).

[8] Gali, J. (1999). *Technology, employment, and the business cycle: do technology shocks explain aggregate fluctuations*. American Economic Review, 89(1), 249–271.

[9] Gali, J., Lopez-Salido, D., and Valles, J. (2007). *Understanding the Effects of Government Spending on Consumption*. Journal of the European Economic Association, 5(1), 227–270.

[10] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.

[11] Collar, F. (2025). *Macroeconomics of Risks - Setting the Stage (Handout 1)*. Lecture notes, Toulouse School of Economics, Fall 2025.

[12] Collar, F. (2025). *Macroeconomics of Risks - Business as Usual (Handout 2)*. Lecture notes, Toulouse School of Economics, Fall 2025.

[13] Collar, F. (2025). *Macroeconomics of Risks - Exogenous Disasters (Handout 3)*. Lecture notes, Toulouse School of Economics, Fall 2025.

[14] Fève, P. (2025). *Macroeconomics of Risks - Shocks and Their Aggregate Effects*. Lecture notes, Toulouse School of Economics, Fall 2025.

[15] Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.

[16] Meddahi, N. (2024). *Time Series Econometrics - ADF Tests*. Course Notes, Toulouse School of Economics.

[17] Smets, F. and Wouters, R. (2007). *Shocks and frictions in U.S. business cycles: a Bayesian DSGE approach*. American Economic Review, 97(3), 586–606.

Appendix

Variable	Description
ngdp	Nominal GDP
ncnd	Nominal nondurable consumption
ncsv	Nominal services consumption
ngov	Nominal government purchases (federal + state + local)
pgdp	GDP deflator
pcndsv	Deflator for consumption nondurables plus services
pgov	Deflator for government purchases

Table 2: List of selected economic and financial variables

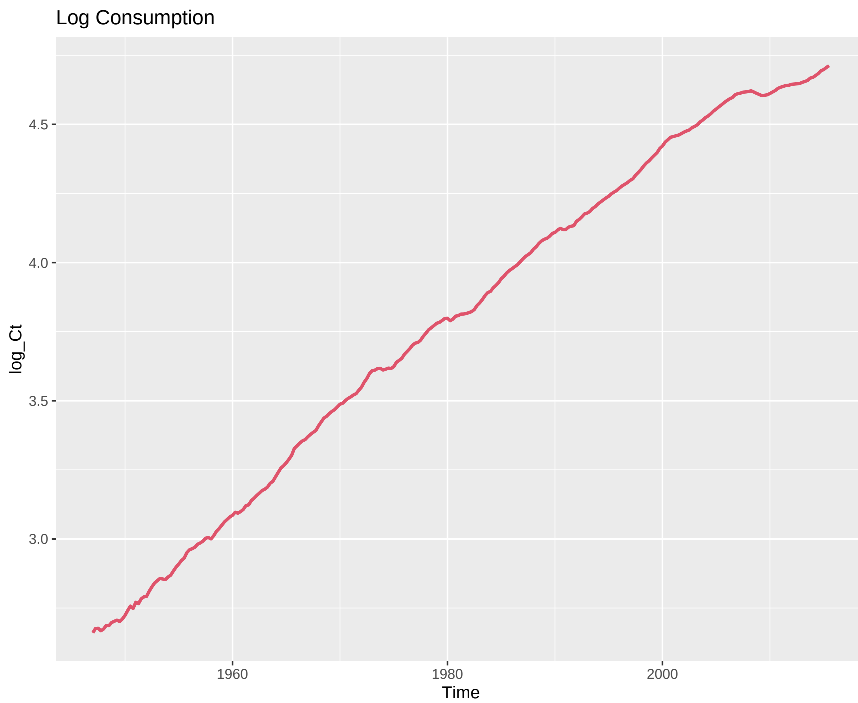

Figure 1: Consumption (logarithm) across time.

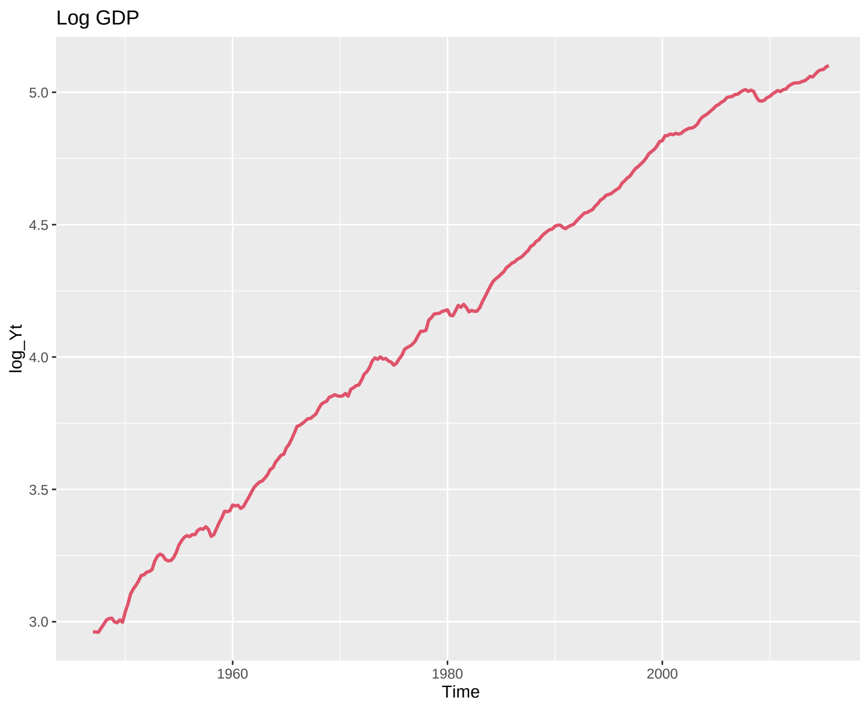

Figure 2: GDP (logarithm) across time.

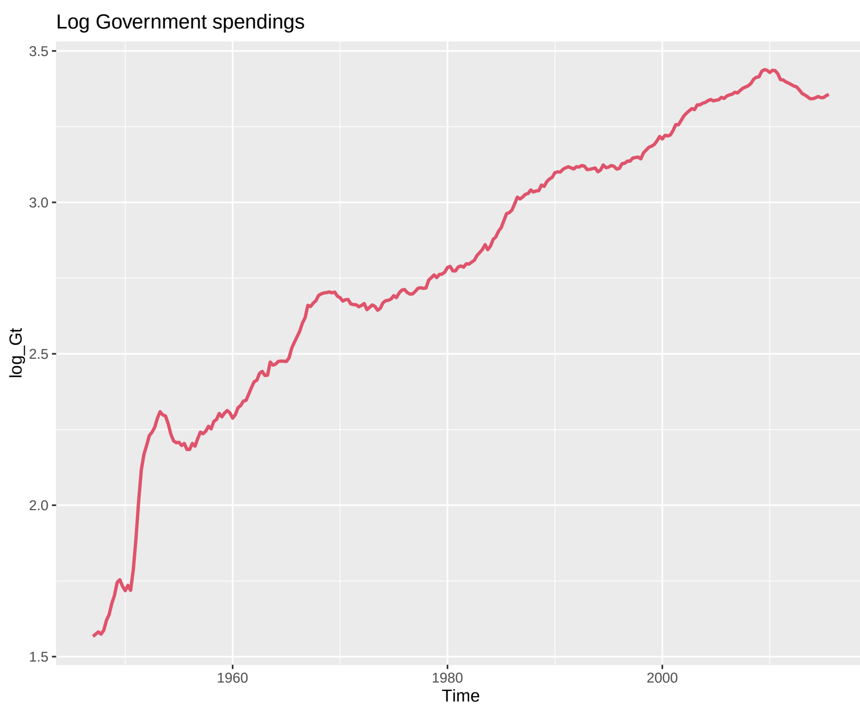

Figure 3: Government spending (logarithm) across time.

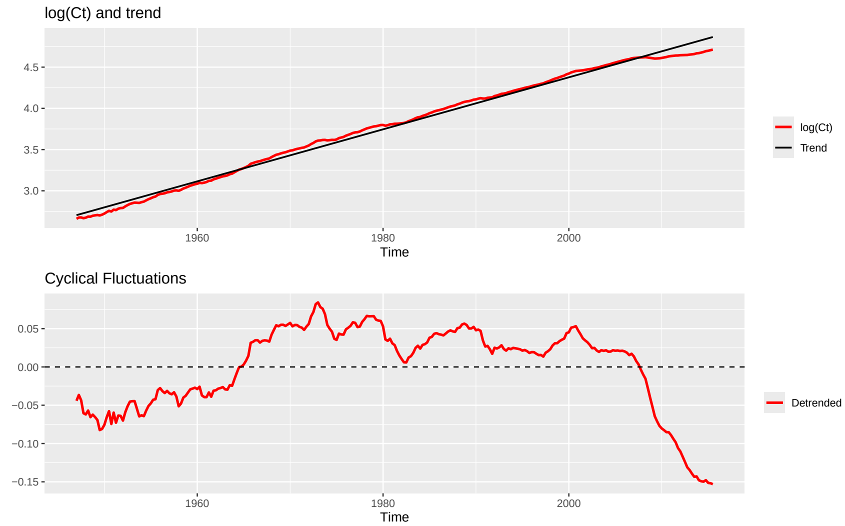

Figure 4: Consumption trend and cyclical fluctuations.

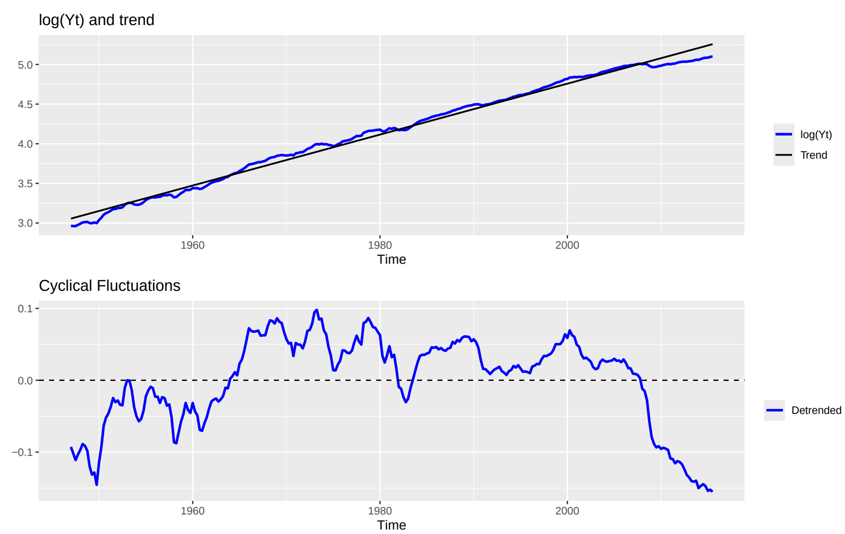

Figure 5: GDP trend and cyclical fluctuations.

Figure 6: Government spending trend and cyclical fluctuations.

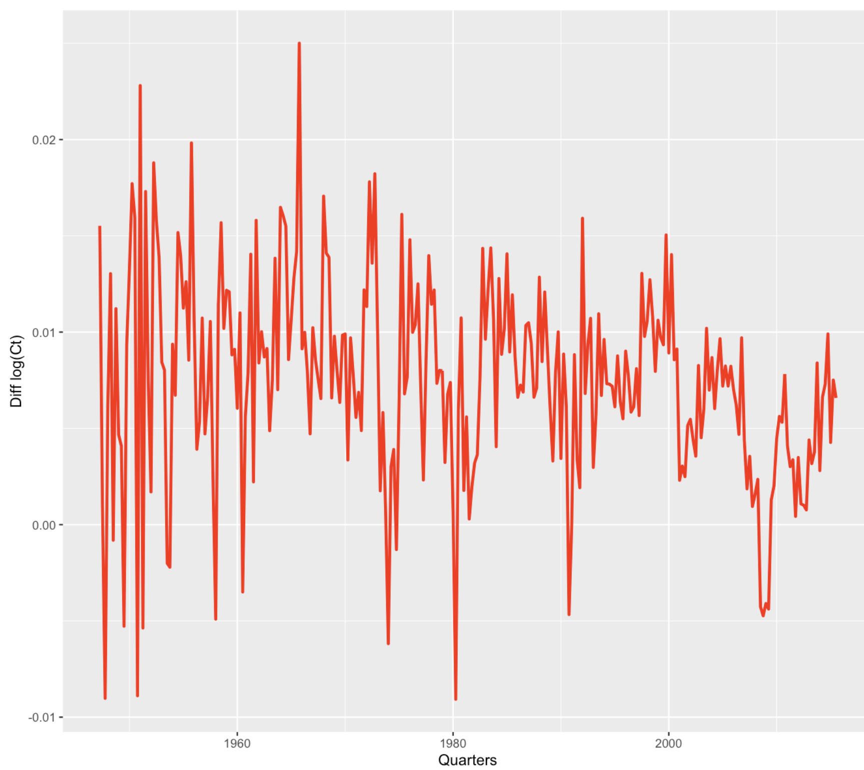

Figure 7: First difference of log consumption.

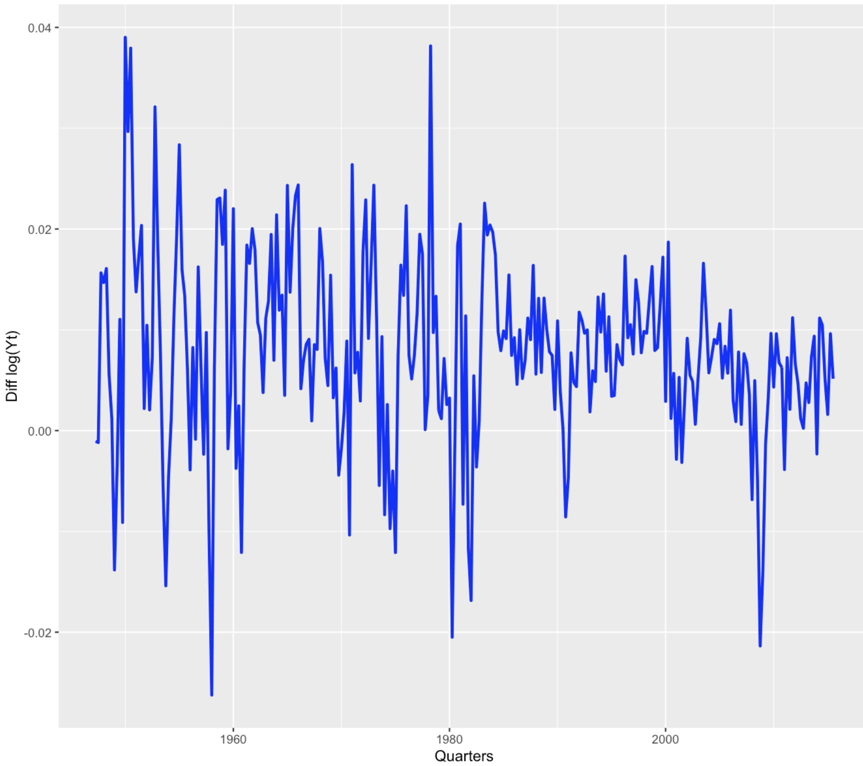

Figure 8: First difference of log GDP.

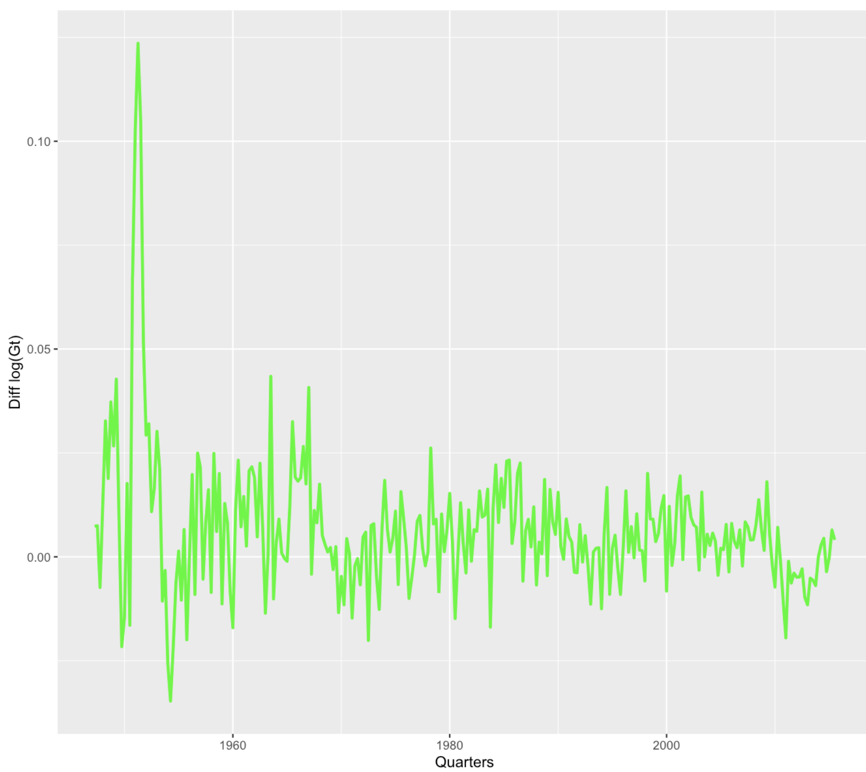

Figure 9: First difference of log public spending.

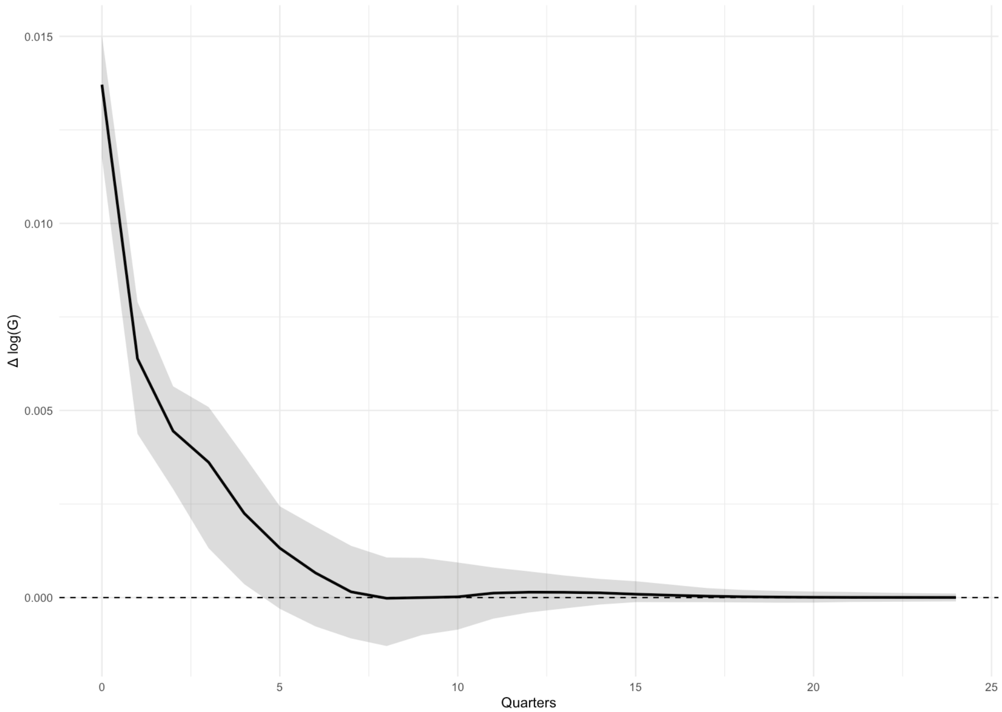

Figure 10: Impulse Response of Δg_t with 95% CI.

Horizon	IRF	Lower 95% CI	Upper 95% CI
1	0.0137	0.0118	0.0150
2	0.0064	0.0044	0.0079
3	0.0045	0.0029	0.0056
4	0.0036	0.0013	0.0051
5	0.0023	0.0004	0.0038
6	0.0013	-0.0003	0.0024
7	0.0007	-0.0008	0.0019
8	0.0002	-0.0011	0.0014
9	-0.0000	-0.0013	0.0011
10	-0.0000	-0.0010	0.0011
11	0.0000	-0.0009	0.0009
12	0.0001	-0.0006	0.0008
13	0.0001	-0.0004	0.0007
14	0.0001	-0.0003	0.0006
15	0.0001	-0.0002	0.0005
16	0.0001	-0.0001	0.0004
17	0.0001	-0.0001	0.0003
18	0.0000	-0.0001	0.0003
19	0.0000	-0.0001	0.0002
20	0.0000	-0.0001	0.0002
21	0.0000	-0.0001	0.0002
22	0.0000	-0.0001	0.0001
23	0.0000	-0.0001	0.0001
24	0.0000	-0.0001	0.0001

Table 3: Impulse Response of Δg_t with 95% CI.

Horizon	Δg_t	Δy_t	Δc_t	i_t
1	1.0000	0.0000	0.0000	0.0000
2	0.9956	0.0038	0.0005	0.0001
3	0.9891	0.0092	0.0007	0.0009
4	0.9737	0.0230	0.0016	0.0017
5	0.9349	0.0617	0.0017	0.0017
6	0.9127	0.0790	0.0066	0.0017
7	0.9000	0.0888	0.0095	0.0017
8	0.8940	0.0910	0.0132	0.0018
9	0.8913	0.0911	0.0156	0.0021
10	0.8902	0.0910	0.0164	0.0024
11	0.8896	0.0909	0.0168	0.0027
12	0.8894	0.0909	0.0169	0.0029
13	0.8892	0.0909	0.0169	0.0030
14	0.8891	0.0909	0.0169	0.0031
15	0.8890	0.0909	0.0169	0.0032
16	0.8888	0.0909	0.0169	0.0033
17	0.8888	0.0909	0.0169	0.0034
18	0.8887	0.0909	0.0169	0.0035
19	0.8886	0.0909	0.0169	0.0036
20	0.8885	0.0909	0.0169	0.0037
21	0.8885	0.0909	0.0169	0.0037
22	0.8884	0.0909	0.0169	0.0038
23	0.8884	0.0909	0.0169	0.0038
24	0.8883	0.0909	0.0169	0.0039

Table 4: FEVD of Δg_t.

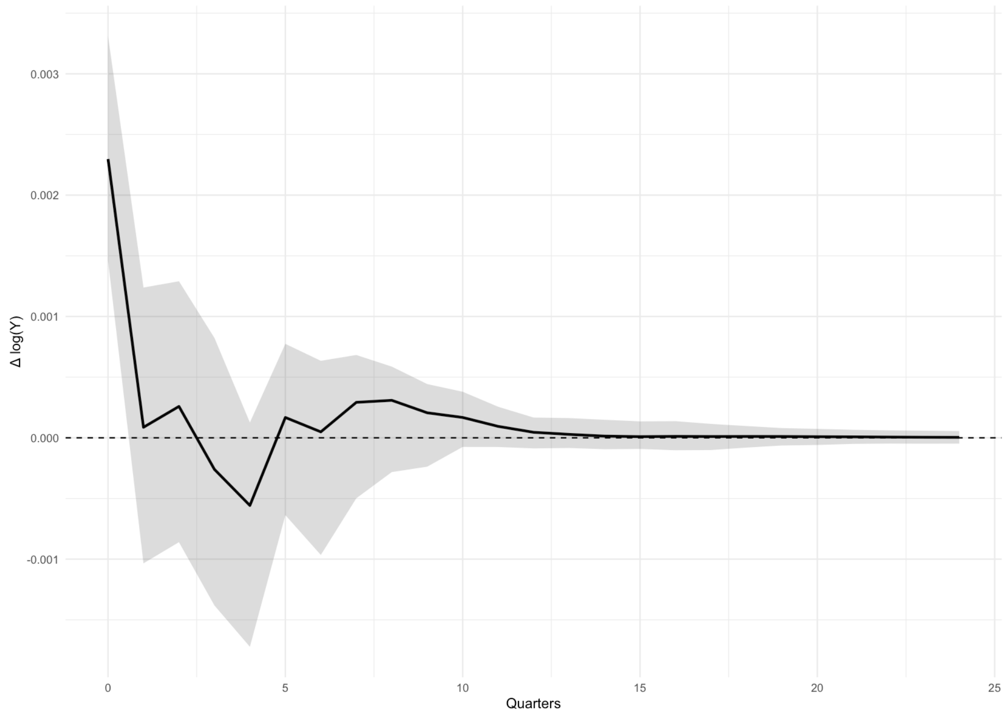

Figure 11: Impulse Response of Δy_t with 95% CI.

Horizon	IRF	Lower 95% CI	Upper 95% CI
1	0.0023	0.0015	0.0033
2	0.0001	-0.0010	0.0012
3	0.0003	-0.0009	0.0013
4	-0.0003	-0.0014	0.0008
5	-0.0006	-0.0017	0.0001
6	0.0002	-0.0006	0.0008
7	0.0000	-0.0010	0.0006
8	0.0003	-0.0005	0.0007
9	0.0003	-0.0003	0.0006
10	0.0002	-0.0002	0.0004
11	0.0002	-0.0001	0.0004
12	0.0001	-0.0001	0.0003
13	0.0000	-0.0001	0.0002
14	0.0000	-0.0001	0.0002
15	0.0000	-0.0001	0.0001
16	0.0000	-0.0001	0.0001
17	0.0000	-0.0001	0.0001
18	0.0000	-0.0001	0.0001
19	0.0000	-0.0001	0.0001
20	0.0000	-0.0001	0.0001
21	0.0000	-0.0001	0.0001
22	0.0000	-0.0001	0.0001
23	0.0000	-0.0001	0.0001
24	0.0000	-0.0001	0.0001

Table 5: Impulse Response of Δy_t with 95% CI.

Horizon	Δg_t	Δy_t	Δc_t	i_t
1	0.0768	0.9232	0.0000	0.0000
2	0.0621	0.8202	0.1010	0.0167
3	0.0578	0.7943	0.1318	0.0162
4	0.0578	0.7838	0.1423	0.0161
5	0.0598	0.7704	0.1540	0.0157
6	0.0600	0.7700	0.1544	0.0157
7	0.0600	0.7698	0.1544	0.0159
8	0.0608	0.7689	0.1543	0.0161
9	0.0617	0.7678	0.1540	0.0165
10	0.0620	0.7672	0.1538	0.0169
11	0.0623	0.7668	0.1537	0.0173
12	0.0623	0.7665	0.1536	0.0175
13	0.0623	0.7663	0.1536	0.0178
14	0.0623	0.7661	0.1536	0.0180
15	0.0623	0.7659	0.1536	0.0182
16	0.0623	0.7658	0.1536	0.0184
17	0.0623	0.7656	0.1536	0.0185
18	0.0623	0.7655	0.1535	0.0187
19	0.0623	0.7654	0.1535	0.0188
20	0.0623	0.7653	0.1535	0.0189
21	0.0622	0.7653	0.1535	0.0190
22	0.0622	0.7652	0.1535	0.0191
23	0.0622	0.7651	0.1535	0.0192
24	0.0622	0.7650	0.1534	0.0193

Table 6: FEVD of Δy_t.

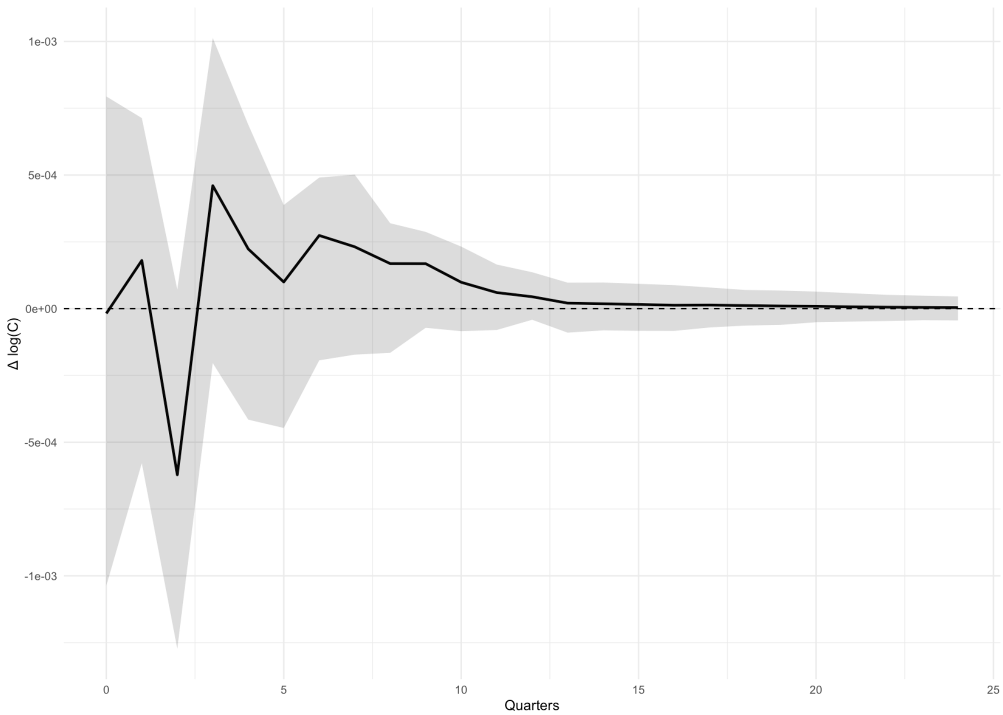

Figure 12: Impulse Response of Δc_t with 95% CI.

Horizon	IRF	Lower 95% CI	Upper 95% CI
1	-0.0000	-0.0010	0.0008
2	0.0002	-0.0006	0.0007
3	-0.0006	-0.0013	0.0001
4	0.0005	-0.0002	0.0010
5	0.0002	-0.0004	0.0007
6	0.0001	-0.0004	0.0004
7	0.0003	-0.0002	0.0005
8	0.0002	-0.0002	0.0005
9	0.0002	-0.0002	0.0003
10	0.0002	-0.0001	0.0003
11	0.0001	-0.0001	0.0002
12	0.0001	-0.0001	0.0002
13	0.0000	-0.0000	0.0001
14	0.0000	-0.0001	0.0001
15	0.0000	-0.0001	0.0001
16	0.0000	-0.0001	0.0001
17	0.0000	-0.0001	0.0001
18	0.0000	-0.0001	0.0001
19	0.0000	-0.0001	0.0001
20	0.0000	-0.0001	0.0001
21	0.0000	-0.0001	0.0001
22	0.0000	-0.0001	0.0001
23	0.0000	-0.0001	0.0001
24	0.0000	-0.0001	0.0001

Table 7: Impulse Response of Δc_t with 95% CI.

Horizon	Δg_t	Δy_t	Δc_t	i_t
1	0.0000	0.2091	0.7909	0.0000
2	0.0014	0.2370	0.7592	0.0024
3	0.0160	0.2456	0.7331	0.0053
4	0.0230	0.2349	0.7356	0.0066
5	0.0246	0.2342	0.7328	0.0084
6	0.0248	0.2329	0.7330	0.0092
7	0.0274	0.2316	0.7296	0.0115
8	0.0292	0.2315	0.7265	0.0128
9	0.0301	0.2312	0.7240	0.0147
10	0.0310	0.2312	0.7215	0.0164
11	0.0312	0.2312	0.7197	0.0179
12	0.0313	0.2310	0.7184	0.0193
13	0.0313	0.2308	0.7175	0.0204
14	0.0313	0.2306	0.7167	0.0214
15	0.0313	0.2304	0.7160	0.0224
16	0.0312	0.2302	0.7154	0.0232
17	0.0312	0.2300	0.7148	0.0240
18	0.0312	0.2299	0.7143	0.0247
19	0.0312	0.2297	0.7138	0.0253
20	0.0312	0.2296	0.7134	0.0259
21	0.0312	0.2294	0.7130	0.0264
22	0.0311	0.2293	0.7126	0.0269
23	0.0311	0.2292	0.7123	0.0274
24	0.0311	0.2291	0.7120	0.0278

Table 8: FEVD of Δc_t.

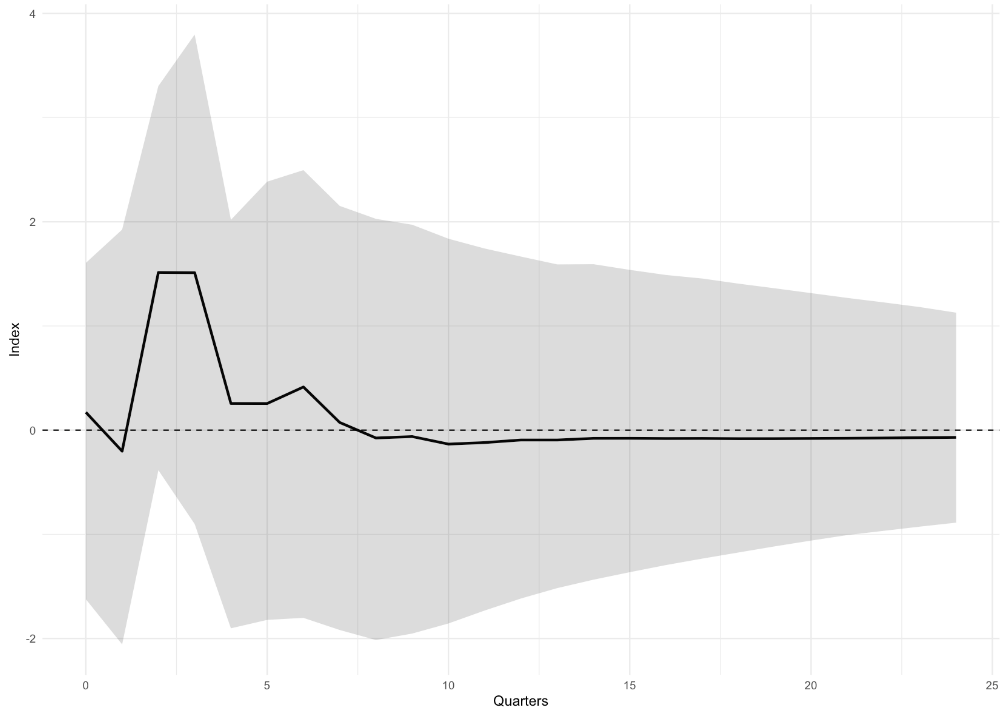

Figure 13: Impulse Response of i_t with 95% CI.

Horizon	IRF	Lower 95% CI	Upper 95% CI
1	0.1713	-1.6230	1.6055
2	-0.2019	-2.0560	1.9250
3	1.5137	-0.3844	3.3033
4	1.5114	-0.9008	3.7966
5	0.2562	-1.9020	2.0188
6	0.2556	-1.8214	2.3851
7	0.4143	-1.8018	2.4958
8	0.0739	-1.9181	2.1526
9	-0.0752	-2.0146	2.0281
10	-0.0616	-1.9525	1.9717
11	-0.1339	-1.8555	1.8372
12	-0.1192	-1.7311	1.7433
13	-0.0945	-1.6158	1.6660
14	-0.0946	-1.5161	1.5909
15	-0.0788	-1.4354	1.5928
16	-0.0783	-1.3633	1.5383
17	-0.0804	-1.2948	1.4893
18	-0.0795	-1.2330	1.4554
19	-0.0816	-1.1740	1.4061
20	-0.0814	-1.1154	1.3615
21	-0.0797	-1.0599	1.3153
22	-0.0780	-1.0072	1.2692
23	-0.0752	-0.9664	1.2263
24	-0.0723	-0.9261	1.1814

Table 9: Impulse Response of i_t with 95% CI.

Horizon	Δg_t	Δy_t	Δc_t	i_t
1	0.0001	0.0000	0.0002	0.9997
2	0.0002	0.0016	0.0002	0.9980
3	0.0060	0.0013	0.0001	0.9925
4	0.0102	0.0012	0.0021	0.9865
5	0.0090	0.0011	0.0024	0.9874
6	0.0083	0.0011	0.0023	0.9883
7	0.0079	0.0012	0.0021	0.9888
8	0.0074	0.0012	0.0020	0.9894
9	0.0070	0.0012	0.0020	0.9899
10	0.0066	0.0011	0.0019	0.9903
11	0.0064	0.0011	0.0018	0.9907
12	0.0061	0.0011	0.0018	0.9910
13	0.0059	0.0011	0.0017	0.9913
14	0.0057	0.0011	0.0017	0.9915
15	0.0056	0.0010	0.0016	0.9917
16	0.0055	0.0010	0.0016	0.9919
17	0.0054	0.0010	0.0016	0.9921
18	0.0053	0.0010	0.0015	0.9922
19	0.0052	0.0010	0.0015	0.9923
20	0.0051	0.0010	0.0015	0.9924
21	0.0050	0.0010	0.0015	0.9925
22	0.0050	0.0010	0.0014	0.9926
23	0.0049	0.0010	0.0014	0.9926
24	0.0049	0.0010	0.0014	0.9927

Table 10: FEVD of i_t.